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I wrote a book: **Introduction to Software Engineering in the Age of AI**. It's free, online, and CC BY 4.0.

Most “AI coding” material lands in one of two camps: prompt tricks that stop working next month, or hand-wringing about whether any of this even counts as engineering. I wanted something in between — software engineering taught as a craft, with AI as a power tool you learn to hold correctly.

So it's built the way you'd actually build: one system, extended chapter by chapter. You start with an empty repository and end with a working “hybrid” mortgage calculator with three front ends — a command line, a graphical UI, and a language model that can operate the whole thing on your behalf. The tools of the trade (terminal, git, Python, a coding agent, local and hosted models) show up exactly when the next chapter needs them, not in a reference dump up front.

The through-line is what I call a **human-verified core with an AI-assisted layer**: you design, write, and verify the core yourself; an agent drafts and a model reasons around it — but neither one gets to decide what “correct” means. That judgment stays with you, written down three times at three altitudes: a plain-language spec, executable tests, and a formal schema a language model can call.

14 chapters, plus tests, CI, and the full build tooling. Everything is in the open.

Read it online: https://rioffe.github.io/intro_to_ai_software_engineering/
Single-file PDF / HTML: https://github.com/rioffe/intro_to_ai_software_engineering/releases/tag/v1.0.0
Source: https://github.com/rioffe/intro_to_ai_software_engineering

Feedback and issues welcome. If you teach this material, it's yours to use.

#SoftwareEngineering #AI #LLM #Programming #OpenEducation #Python